CHE 201 Fall 2019 HW 10

Alex Boskov

TOTAL POINTS

97 / 100

QUESTION 1

1 1 **10 / 10**

✓ **- 0 pts** Correct

- 2 pts (g) incorrect

- 2 pts (f) where is the answer?

QUESTION 2

2 2 **15 / 15**

✓ **- 0 pts** Correct

- 3 pts why QHxWR?

- 8 pts Determine which cycle discharge /transfer greater energy

- 3 pts last summary should be cycle I discharge greater energy

- 3 pts (a) (ii) cycle I discharges greater energy

- 5 pts Definition of A and B, e.g. what is QHA and QHB? Why and What is IQ>RQ?

QUESTION 3

3 3 **15 / 15**

✓ **- 0 pts** Correct

- 5 pts total thermal efficiency incorrect

- 5 pts (a) total thermal efficiency need to be expressed in individual thermal efficiency

- 5 pts T incorrenct

- 5 pts (b) total thermal efficiency?

QUESTION 4

4 4 **15 / 15**

✓ **- 0 pts** Correct

- 3 pts Missing or wrong calculation of max thermal efficiency (66.7%)

- 1.5 pts a) efficiency = 0.5

- 1.5 pts a) Irreversible

- 3 pts b) reversible

- 3 pts c) irreversible

- 1.5 pts c) efficiency = 0.6

- 3 pts d) impossible

- 9 pts Missing calculation of thermal efficiency for a, b and c

- 15 pts No answer

QUESTION 5

5 5 **15 / 15**

✓ **- 0 pts** Correct

- 6 pts Missing part b

- 3 pts Did not calculate the intermediate temperature T

- 4.5 pts Did not calculate any of the efficiencies

- 3 pts Did not calculate Work for a single cycle

- 15 pts No answer

QUESTION 6

6 6 **14 / 15**

- 0 pts Correct

✓ **- 1 pts a) Qin = 3.496 x 10^7 kJ (not 10^6)**

- 1 pts a) Calculation error. Qin = 3.496 x 10^7 kJ

- 5 pts a) Wrong Qin

- 5 pts c) max coeff = 24.5

- 15 pts No answer

QUESTION 7

7 7 **13 / 15**

- 0 pts Correct

- 2 pts c) Did not comment on the result

- 2 pts σ cycle = ?

- 6 pts Missing or wrong answer to part c

- 2 pts σ cycle = 0.5 kJ/(kg.K)

✓ **- 2 pts Missing σ cycle unit (kJ/(kg*k))**

- 3 pts Missing answer to part b

- 15 pts No answer

HW 10

3 3

Bosko

1. [entropy change] = [entropy transfer] + [energy production]

(a) possible (b) possible (c) impossible (d) indeterminate

(e) possible (f) impossible (g) indeterminate

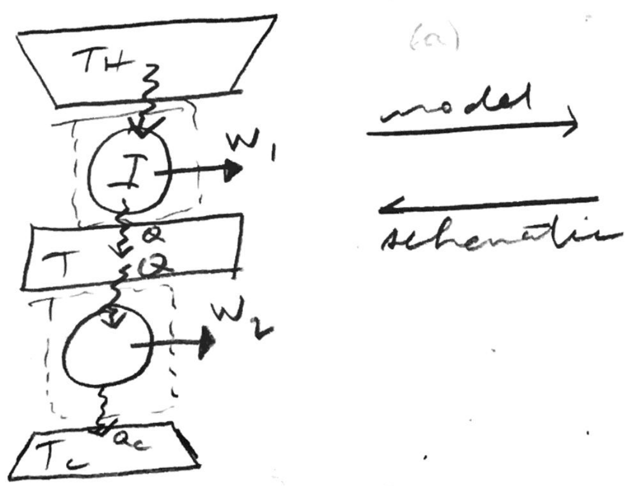
2. (a) (i) $\boxed{R}$ develops greater net work, because it is less efficient thermally $\left(\eta = \frac{W}{Q_H}\right)$

(ii) $\boxed{I}$ discharges greater energy by heat transfer to the cold reservoir because it produces less work

(b) (i) $\boxed{I}$ receives greater energy by heat transfer because it produces same net work as I but is less thermally efficient

(ii) $\boxed{I}$ discharges greater energy by heat transfer because it is less thermally efficient than R

3.



. energy transfers are equal and positive

(a) $\eta_{overall} = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H}$

$$= 1 - \left(\frac{Q_C}{Q}\right)\left(\frac{Q}{Q_H}\right)$$

$$= 1 - (1-\eta_1)(1-\eta_2)$$

$$\boxed{\eta_{overall} = (\eta_1 + \eta_2) - (\eta_1 \eta_2)}$$

OVER →

1 1 **10 / 10**

✓ **- 0 pts** Correct

- 2 pts (g) incorrect

- 2 pts (f) where is the answer?

HW 10 3 3 Bosko

1. [entropy change] = [entropy transfer] + [energy production]
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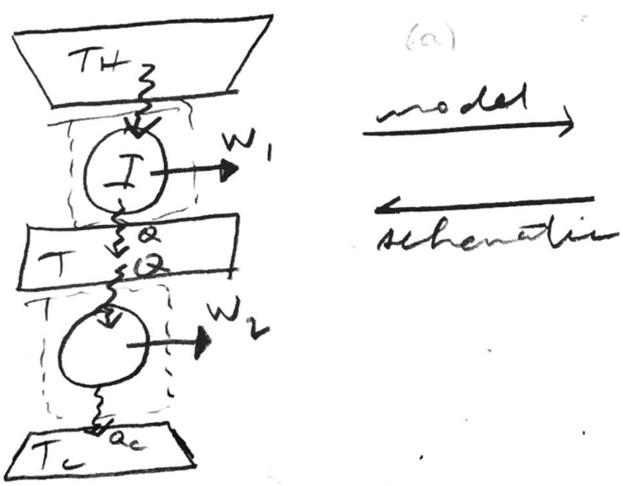

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$= 1 - (1-\eta_1)(1-\eta_2)$

$\boxed{\eta_{overall} = (\eta_1 + \eta_2) - (\eta_1\eta_2)}$

OVER →

2 2 15 / 15

- ✓ **- 0 pts** Correct
- **- 3 pts** why QHxWR?
- **- 8 pts** Determine which cycle discharge /transfer greater energy
- **- 3 pts** last summary should be cycle I discharge greater energy
- **- 3 pts** (a) (ii) cycle I discharges greater energy
- **- 5 pts** Definition of A and B, e.g. what is QHA and QHB? Why and What is IQ>RQ?

HW 10 3 3 Bosko

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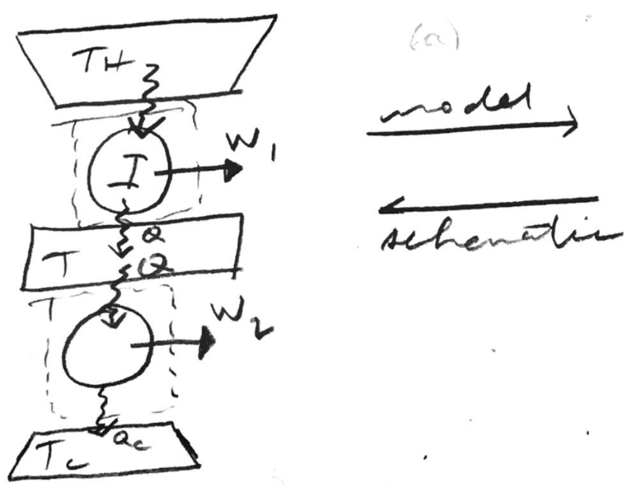

(a)

- energy transfers are equal and positive

(a)
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$$= 1 - \left(\frac{Q_c}{Q}\right)\left(\frac{Q}{Q_H}\right)$$
$$= 1 - (1-\eta_1)(1-\eta_2)$$
$$\boxed{\eta_{overall} = (\eta_1 + \eta_2) - (\eta_1 \eta_2)}$$

OVER →

BOSKOV HW (2)

(b) $\eta_{overall} = (\eta_1+\eta_2) - (\eta_1\eta_2)$

$$= \left(1-\frac{T}{T_H} + 1 - \frac{T_C}{T}\right) - \left(1-\frac{T}{T_H}\right)\left(1-\frac{T_C}{T}\right)$$

$$= \left(2 - \frac{T}{T_H} - \frac{T_C}{T}\right) - \left(1 - \frac{T_C}{T} - \frac{T}{T_H} + \frac{TT_C}{TT_H}\right)$$

$$= \boxed{1 - \frac{T_C}{T_H}}$$

because the processes are reversible, overall efficiency involves only endpoints

(c) $\eta_1 = \eta_2$

$$\frac{T}{T_H} = \frac{T_C}{T} \rightarrow T^2 = T_C T_H \rightarrow \boxed{T = \sqrt{T_C T_H}}$$

4. $\eta = 1 - \frac{Q_C}{Q_H} = \frac{W}{Q_H} \rightarrow$ equal to $\frac{T_C}{T_H}$ if reversible

(a) $\frac{450}{900} = .5$ $\quad 1 - \frac{400}{1200} = .67$ $\quad$ operates irreversibly

(b) $1 - \frac{300}{900} = .67$ $\quad 1 - \frac{400}{1200} = .67$ $\quad$ operates reversibly

(c) $Q_H = W + Q_C = 600 + 400 = 1000$ Btu

$\eta = \frac{600}{1000} = .6$ $\quad$ operates irreversibly

(d) $\eta = .75$ $\quad 1 - \frac{400}{1200} = .67$ $\quad$ impossible, since $\eta > .67$

3 3 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 5 pts** total thermal efficiency incorrect
- **- 5 pts** (a) total thermal efficiency need to be expressed in individual thermal efficiency
- **- 5 pts** T incorrenct
- **- 5 pts** (b) total thermal efficiency?

BOSKOV HW x (2)

(b) $\eta_{overall} = (\eta_1+\eta_2)-(\eta_1\eta_2)$

$$= \left(1-\frac{T}{T_H}+1-\frac{T_c}{T}\right)-\left(1-\frac{T}{T_H}\right)\left(1-\frac{T_c}{T}\right)$$

$$= \left(2-\frac{T}{T_H}-\frac{T_c}{T}\right)-\left(1-\frac{T_c}{T}-\frac{T}{T_H}+\frac{TT_c}{TT_H}\right)$$

$$= \boxed{1-\frac{T_c}{T_H}}$$

because the processes are reversible, overall efficiency involves only endpoints

(c) $\eta_1 = \eta_2$

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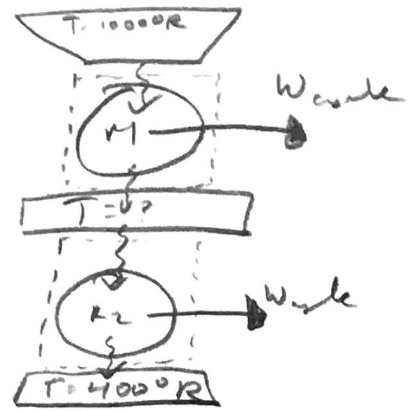
4 4 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 3 pts** Missing or wrong calculation of max thermal efficiency (66.7%)
- **- 1.5 pts** a) efficiency = 0.5
- **- 1.5 pts** a) Irreversible
- **- 3 pts** b) reversible
- **- 3 pts** c) irreversible
- **- 1.5 pts** c) efficiency = 0.6
- **- 3 pts** d) impossible
- **- 9 pts** Missing calculation of thermal efficiency for a, b and c
- **- 15 pts** No answer

5.

BOSKOV Alex 13



model

schematic

- cycles are reversible
- all energy transfer in dir. of arrows are positive

(a) $T_C - T = W_{cycle} = T - T_H$

$2T = T_C + T_H$

$\boxed{T = 700°R}$

$\eta_1 = 1 - \frac{T}{T_H} = 1 - \frac{700}{1000} = \boxed{.3}$

$\eta_2 = 1 - \frac{T_C}{T} = 1 - \frac{400}{700} = \boxed{.43}$

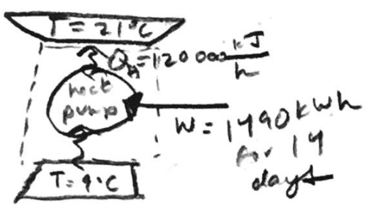
(b) $\eta = 1 - \frac{400}{1000} = \boxed{.6}$

$W_{cycle} = Q_H - Q_C$

$= (Q_H - Q) + (Q - Q_C)$

$= \boxed{2 W_{cycle}}$

6.



model

schematic

- steady-state
- power in form of electricity

(a) $1490 \text{ kWh} \times \frac{1 \text{ kJ/s}}{1 \text{ kW}} \times \frac{60 \text{ min}}{1 \text{ hour}} \times \frac{60 \text{ sec}}{1 \text{ min}} = 5364000 \text{ kJ}$

$\frac{120000 \text{ kJ}}{1 \text{ hr}} \times \frac{24 \text{ hr}}{1 \text{ day}} (14 \text{ days}) = 40320000$

$536400 \text{ kJ} - 40320000 \text{ kJ} = \boxed{3495600 \text{ kJ}}$

(b) $\frac{5364000 \text{ kJ}}{14 \text{ days}} \times \frac{1 \text{ day}}{24 \text{ hours}} = 15964.3 \frac{\text{kJ}}{\text{h}}$

$\gamma = \frac{Q_H}{W_{cycle}} = \frac{120000}{15964.3} = \boxed{7.52}$

(c) $21 + 273 = 294 \text{ K}$

$9 + 273 = 282 \text{ K}$

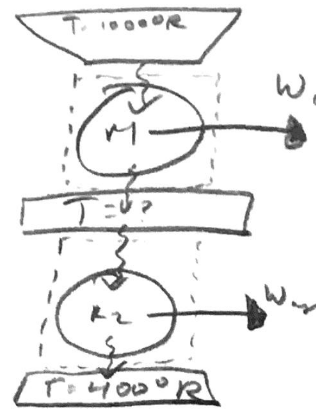
$\gamma = \frac{T_H}{T_H - T_L} = \frac{294}{294 - 282} = \boxed{24.5}$

5 5 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 6 pts** Missing part b
- **- 3 pts** Did not calculate the intermediate temperature T
- **- 4.5 pts** Did not calculate any of the efficiencies
- **- 3 pts** Did not calculate Work for a single cycle
- **- 15 pts** No answer

BOSKOV Alex 13

5.



model →

← schematic

- cycles are reversible
- all energy transfer in dir. of arrows are positive

(a) $T_C - T = W_{cycle} = T - T_H$

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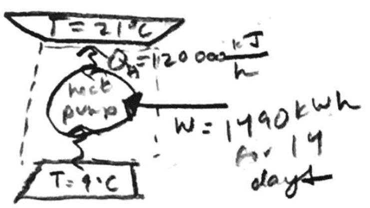

model →

← schematic

- steady-state
- power in form of electricity

(a) $1490\,kWh \times \frac{1\,kJ/s}{1\,kW} \times \frac{60\,min}{1\,hour} \times \frac{60\,sec}{1\,min} = 5364000\,kJ$ $\quad \frac{120000\,kJ}{1\,hr} \times \frac{24\,hr}{1\,day}(14\,days) = 40320000$

$5364000\,kJ - 40320000\,kJ = \boxed{3495600\,kJ}$

(b) $\frac{5364000\,kJ}{14\,days} \times \frac{1\,day}{24\,hours} = 15964.3 \frac{kJ}{h}$

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$9 + 273 = 282\,K$

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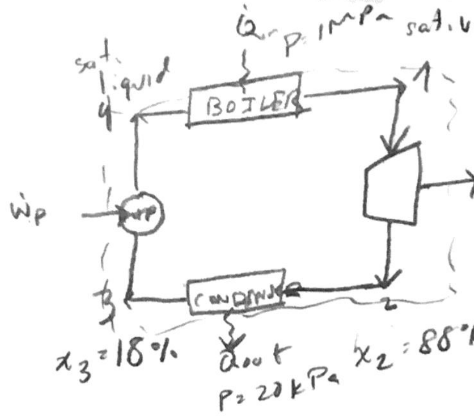
6 6 **14 / 15**

- **- 0 pts** Correct
- ✓ **- 1 pts a) Qin = 3.496 x 10^7 kJ (not 10^6)**
- **- 1 pts** a) Calculation error. Qin = 3.496 x 10^7 kJ
- **- 5 pts** a) Wrong Qin
- **- 5 pts** c) max coeff = 24.5
- **- 15 pts** No answer

BOSCOO

7.



model → · boiler, condenser at constant pressure

turbine and pump adiabatic

← schematic · KE and PE can be ignored

steady-~~state~~

· control volumes

· each has 1 inlet 1 outlet

(a) $$\oint\left(\frac{\delta Q/\dot m}{T}\right)_b = -\frac{\sigma_{cycle}}{\dot m} = \frac{Q_H/\dot m}{T_H} - \frac{Q_C/\dot m}{T_C}$$

$T_{41} \overset{1MPa}{\Longrightarrow} 179.9°C + 273 = 452.9 K$

$T_{23} \overset{20kPa}{\Longrightarrow} 60.06°C + 273 = 333.06 K$

$\dot Q_{in} = \dot m(h_1 - h_4) = 2015.3\dot m$

$h_1 \overset{A-3}{\Longrightarrow} 2778.1 \ kJ/kg \qquad h_4 \overset{A-3}{\Longrightarrow} 762.81 \ kJ/kg$

$\dot Q_{out} = \dot m(h_2 - h_3) = 1648.16\dot m$

$h_2 \overset{A-3}{\Longrightarrow} (1-0.88)(251.4) + (0.88)(2609.7) = 2324.06 \ \frac{kJ}{kg}$

$h_3 \overset{A-3}{\Longrightarrow} (1-0.18)(251.4) + (.18)(2609.7) = 675.9 \ kJ/kg$

$$-\frac{\sigma_{cycle}}{\dot m} = \frac{2015.3}{452.9} - \frac{1648.16}{333.06} \rightarrow \sigma_{cycle} = 0.5$$

$\sigma_{cycle} = 0.5 \geq 0$, so the cycle is $\boxed{\text{irreversible}}$

(b) $$\eta = 1 - \frac{\dot Q_{out}}{\dot Q_{in}} = 1 - \frac{1648.16}{2015.3} = \boxed{0.18}$$

(c) $$\eta_c = 1 - \frac{T_{32}}{T_{41}} = 1 - \frac{333.06}{452.9} = \boxed{0.264}$$

the Carnot efficiency is greater than actual because irreversibilities exist in the ~~process~~

7 7 **13 / 15**

- **- 0 pts** Correct
- **- 2 pts** c) Did not comment on the result
- **- 2 pts** σ cycle = ?
- **- 6 pts** Missing or wrong answer to part c
- **- 2 pts** σ cycle = 0.5 kJ/(kg.K)
- ✓ **- 2 pts Missing σ cycle unit (kJ/(kg*k))**
- **- 3 pts** Missing answer to part b
- **- 15 pts** No answer